

components and improves quality of soldering and desoldering.

Tips

In addition to temperature control of a soldering iron, some other factors can also affect the heat transfer and control, such as the size and shape of the tip, the type and thickness of the material being soldered, and the amount of solder being used. Proper heat control is essential for achieving a successful and reliable solder joint.

Companies offer a variety of soldering tips that enable precision while working. These tips vary as per their applications, like hoof tips are meant for larger amounts of solder, knife tips to reach into the slotted cavities and edges, chisel tips are multi-purpose solder tips, and conical tips are for precision work. Companies like JBC and Weller offer a broad range of tips to choose from.

If you are buying tips for an existing soldering station, make sure that the tips are built to match and operate with your equipment. Some tips are intended for heavy-duty use while the others are intended for light-duty and precision work.

Sleep feature

Tip life increases exponentially by using lower temperatures. Using the sleep mode, the average temperature can be further reduced, which can extend a tip's life five times. The sleep mode automatically lowers tip temperature below solder melting point when the tool rests in its stand.

Factors Affecting Tip Life

Incorrect technique. Too much pressure on the surface to solder can cause tip or plate breakage. Do not use it as a pointed object or a screwdriver. Choosing the biggest tip possible for each application and working at the lowest possible temperature might do the thing.

Iron oxide does not wet nor transfer temperature well (dewetting). The higher the temperature, the greater the risk of oxidation. Keep the tips always tinned while not in use and keep them clean.

Too much active flux, causing corrosion. IPC recommends ROL, REL, and ORL flux. The internal flux from small Ø solder reels might not be enough. Providing compatible external flux might help.

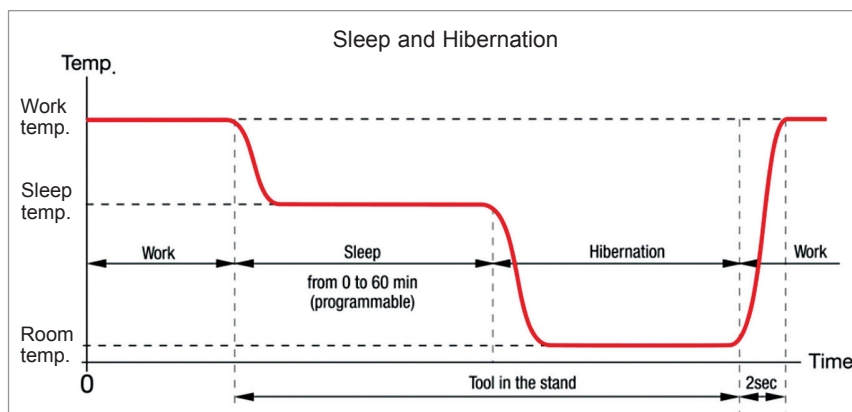


Fig. 2: Graph representing the working of sleep mode of soldering iron (Credit: JBC Tools)

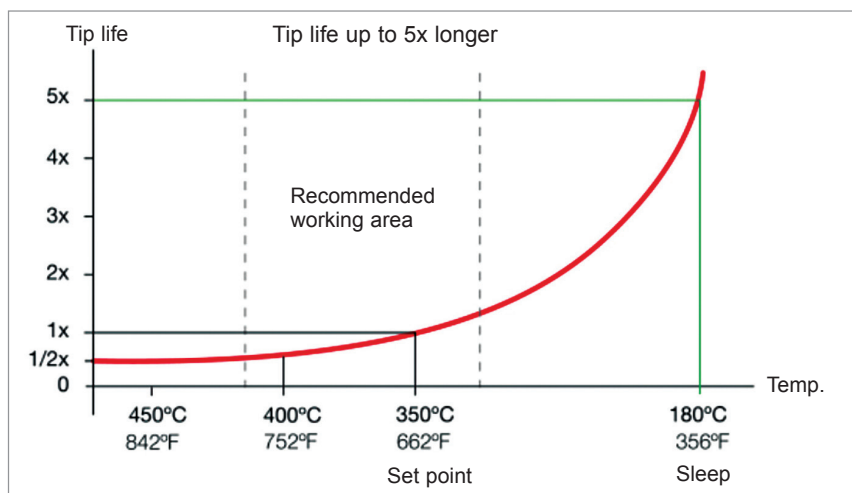


Fig. 3: Graph showing increase in life of the product w.r.t. to temperature (Credit: JBC Tools)

It prevents the dissolution of the tip's iron coating into molten solder.

After a configurable period of tool inactivity in the stand (10 minutes factory default), the tool enters into hibernation mode, which cuts off the power supply and lets the tip reach room temperature, thus preventing oxidation, and saving en-

ergy. Some of the products that offer this feature are WE 1010 soldering iron from Weller Tools, JBC's B-iron, and PX-280 by Goot (Taiyo Electric Ind. Co.).

Soldering stations

Soldering station is the other half of the soldering iron. Soldering stations enable working efficiency and productivity in multiple ways. These are generally used to keep the soldering irons between their use.

But there can also be the need to simultaneously use different soldering tips. To enable such transition, soldering stations are designed to replace the tools needed for changing tips.

JBC's soldering station comes with a quick cartridge extractor at-